

NHS data analysis report

The data analysis process started with reviewing the documentation and provided datasets it is to say uh exploring the metadata and three data sets provided.

The goal of the analysis is to try to answer the following questions.

Has there been adequate staff and capacity in the networks?

What was the actual utilization of resources?

To answer these questions, I analyzed the appointment distribution across ICB locations and took deeper look at appointments from different perspectives such as

What was the distribution of appointments per national category HCP type, context type
Seasonal trends in appointments.

Analyzed missed appointments to determine whether there is a pattern on “why people skip medical appointments?”

Jupyter Notebook section.

I started the process with imports and all the data sets in PyCharm and since checking the data this included checking datasets heads looking if there are missing values and also checking data types to make sure they will fit for the analysis process.

After that I checked data ranges in all three data sets to see if they cover the same Time.

When the data validation was finished, I decided to look at some common things like locations with the most appointments, the most popular national categories context types and service settings. When this was done, I determined top five locations with the most appointments. Though the Jupiter notebook does not have these visualization it looks like top five ICB locations differ from the rest drastically. The number of appointments increases rapidly in comparison to other locations.

The next step in my analysis what start creating visualizations for representing the findings.

For these purposes I utilized matplotlib and Seaborn packages. While creating visuals I try to make them big enough to be readable inside the IDE. Also, I tried to use color palette with high contrast so that each value on chart it's visible distinctively. Visualizations also contained the legend for easier interpretation.

Visualization section contains some short conclusions drawn from the charts and represents my own observations and interpretation of the data. The brief conclusions were subsequently elaborated in greater detail in a PowerPoint presentation.

I also tried to analyze the available Twitter data to see if there is any trend or any valuable insight can be drawn from it the Twitter data was parsed according to the hashtag it contains. Also, I wanted to look at the most liked popular and retweeted Tweets. As per this analysis I have not found anything useful personally for me. That's why I can draw the conclusion that this data is not relevant. If we are interested in what people say about NHS, we should scrap the Twitter data more precisely and probably compose a list of hashtags, that's what we'll use in future web scraping. It is also particularly important to note that the geographical region plays a significant role. Finally, to say about the Twitter data is that at first glance it looks like streets are related to some corporate and business accounts while the NHS should concentrate probably on monitoring the general public sector. By that I mean the tweets from usual people across the UK. I assume that scraping this amount of data will require significant computational resources so I think at first they should do a test run and see if this Twitter data is really important to spend money on in future.

I chose to further examine missed appointments marked as DNA in the dataset. I decided to concentrate on missed appointments because this is one of the main reasons for budget loss for NHS.

As per NHS in for public information the missed appointments cost around £600 million in the last year. So, analyzing missed appointments can help you to understand the following:

- What types of appointments are skipped most of all?
- Can we utilize the resources in any other way to decrease this number and save money?
- Is there any link between type of appointment, HCP type and skipped appointment?

I filtered out all the missed appointments from the document and started analyzing them from the perspectives mentioned above. I think it's especially important to understand this particular NHS problem because it affects lots of things such as budget workload the number of personnel and the accessibility of public health care services for other people.

I discovered several interesting patterns and missed appointments and uh they will be represented in my presentation.

Recommendations regarding how data collection and analytic efforts could be improved.

While doing the initial data check I noticed a remarkably interesting thing. Many categories and values have attributes such as unknown or unmapped. Considering the fact that GP practices across the UK use different systems for medical patient records and appointment management, I assume the disinformation is later sent to NHS in a form of report.

Though I have not seen any of the systems and unaware of any standards and procedures in place for data collection I would recommend that

NHS should revise its standards for data collection and develop clear instructions on what they need from ICB's and GP practices from the reporting perspective.

Assuming the fact that public health care is a regulated environment, I will also suggest that NHS or ICBs start demanding software providers follow these guidelines in terms of data collection. By that I mean the software companies must adjust their systems to allow them to collect the data in clear format so that NHS can perform data analysis and make more precise decisions in terms of hiring new people or rebalancing the current resources and budgets.

During work I also used git for saving the results of my work. The link to the repository is included, it is at the bottom of the Jupiter notebook.